Quality, Reliability & Maintenance ME3327E

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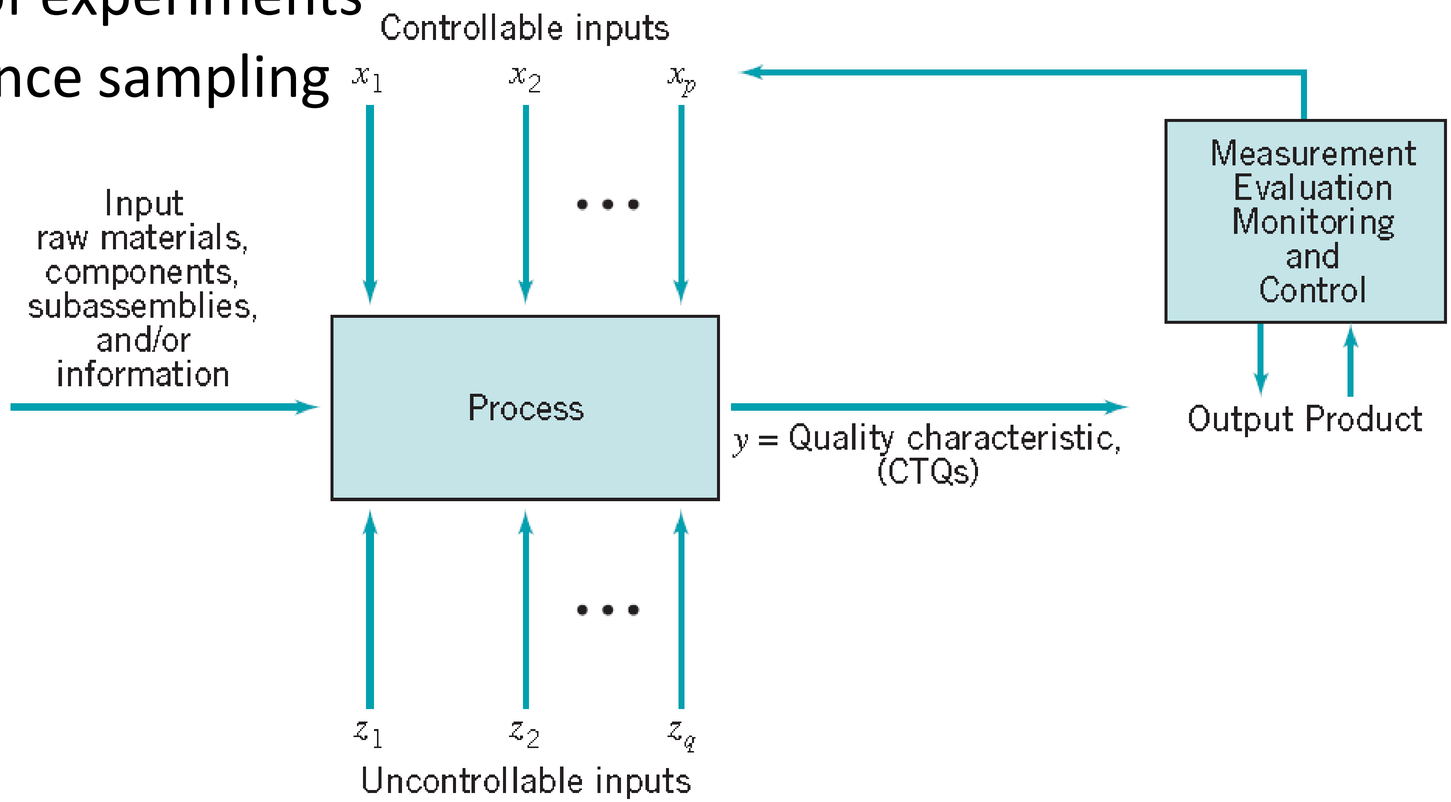
Statistical Methods for Quality Control and Improvement

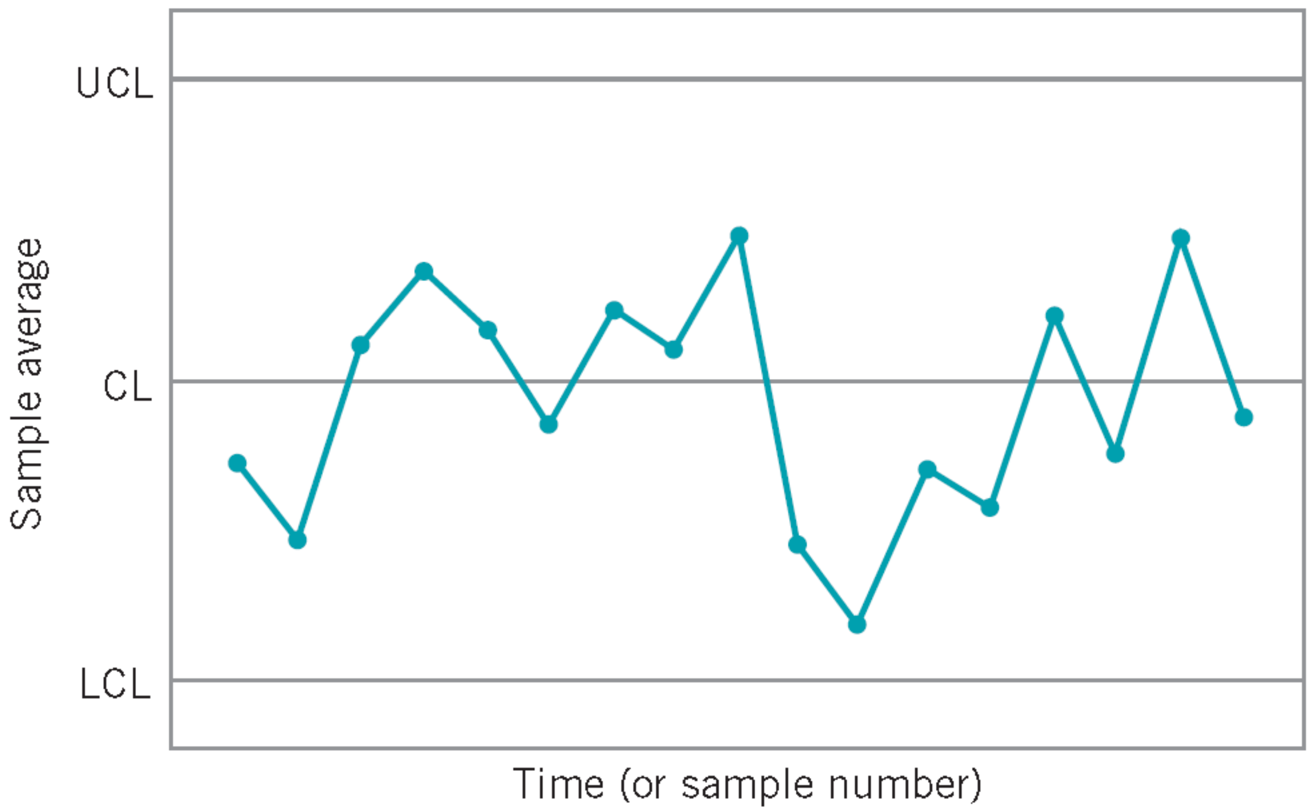
- Three major areas:
  - Statistical process control
  - Design of experiments
  - Acceptance sampling



Production process inputs and outputs

Statistical process control

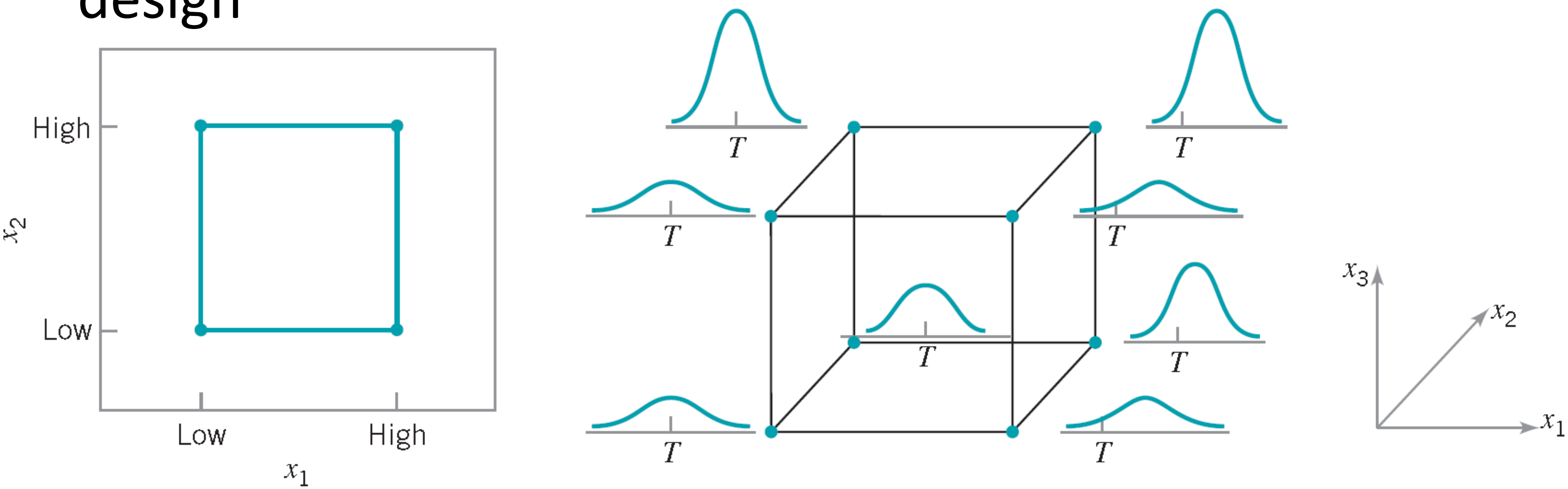
- A control chart is one of the primary techniques of statistical process control (SPC).



A typical control chart

Design of experiments

- Genichi Taguchi
- One major type of designed experiment is the factorial design

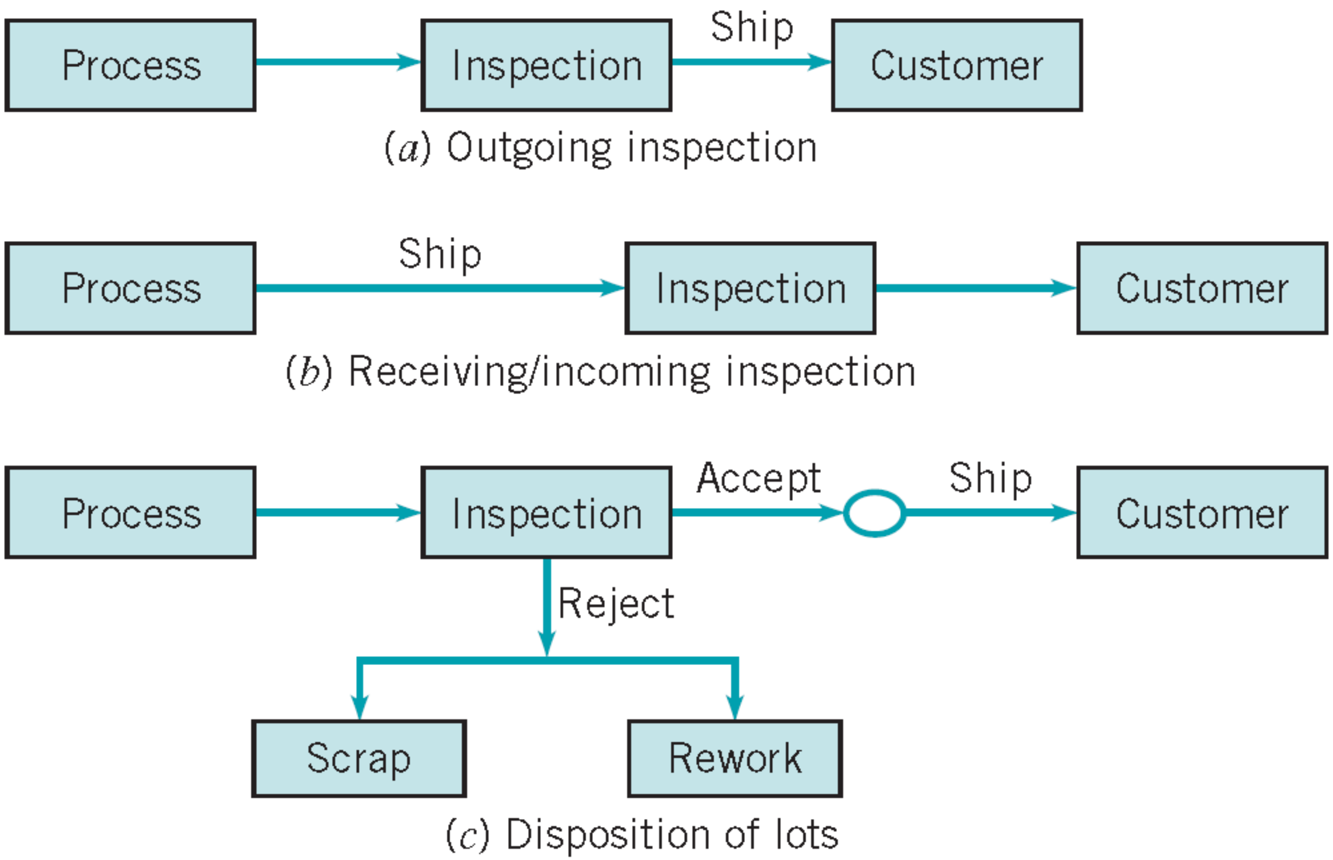


(a) Two factors, x_1 and x_2 (b) Three factors, x_1, x_2, and x_3

Factorial designs for the process

Acceptance sampling

Acceptance sampling, defined as the inspection and classification of a sample of units selected at random from a larger batch or lot and the ultimate decision about disposition of the lot, usually occurs at two points: incoming raw materials or components, or final production.



Variations of acceptance sampling

Other basic tools for quality improvement

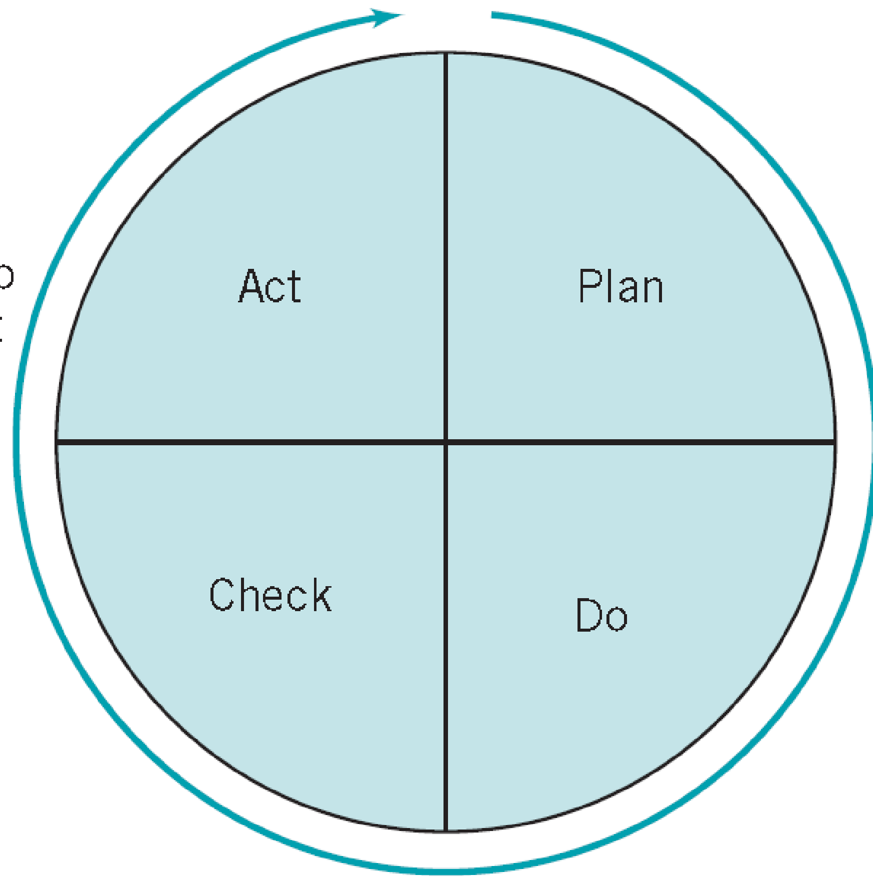
- Flowcharting of processes
- Pareto analysis
- Cause-and-effect diagram
- Histogram
- Control charts
- Check sheets
- Scatter plots

There are several versions of this problem-solving methodology, each of which combines or splits some of the above steps. Dr. Deming's "plan-do-check-act" (PDCA) cycle and Dr. Juran's "breakthrough sequence" are examples of problem-solving approaches.

Deming's PDCA Cycle

Adopt the change or abondon it. If adopted, make sure that it leads to permanent improvement

Plan a change or an experiment aimed at system improvement



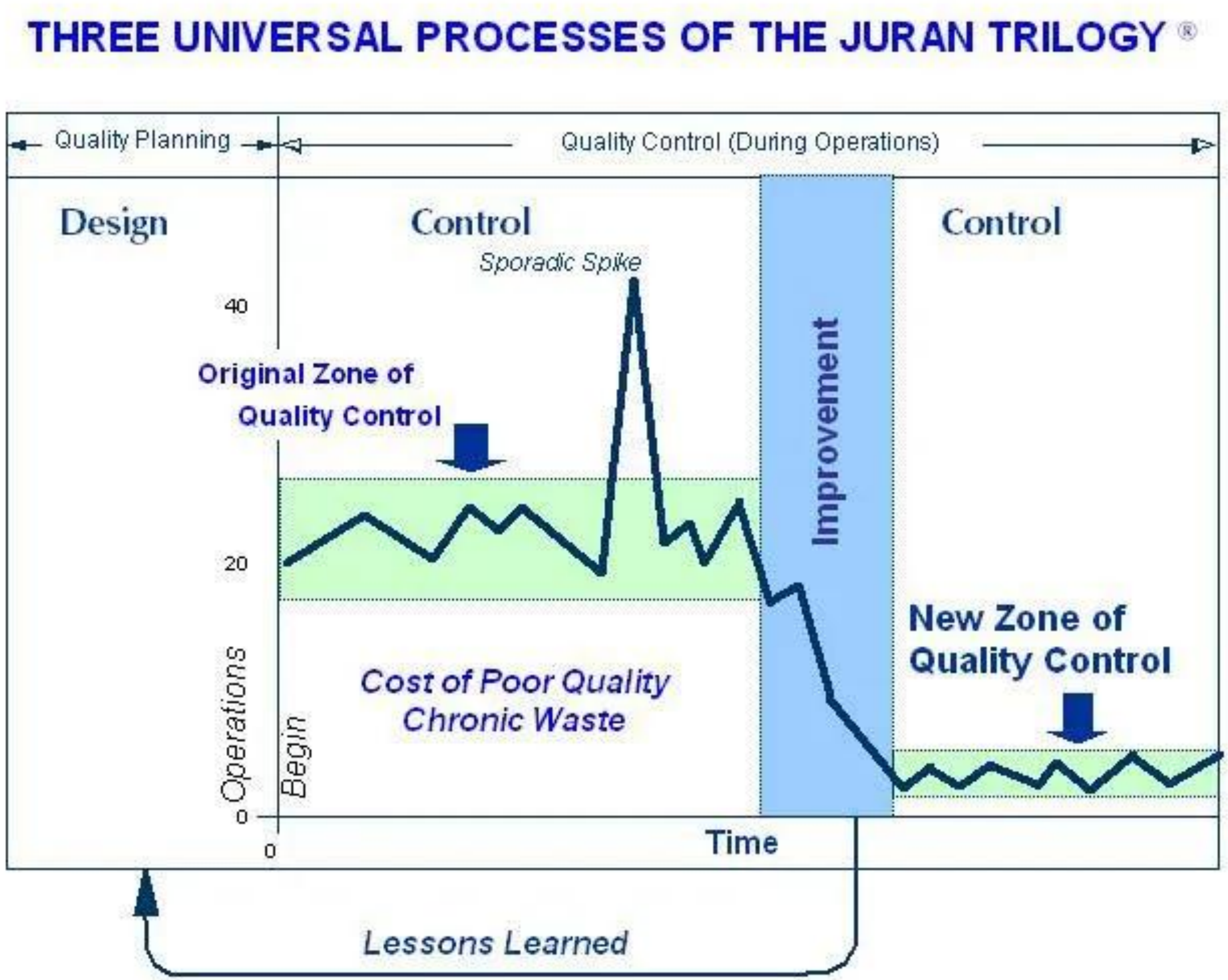
Study and analyze the results obtained. What was learned?

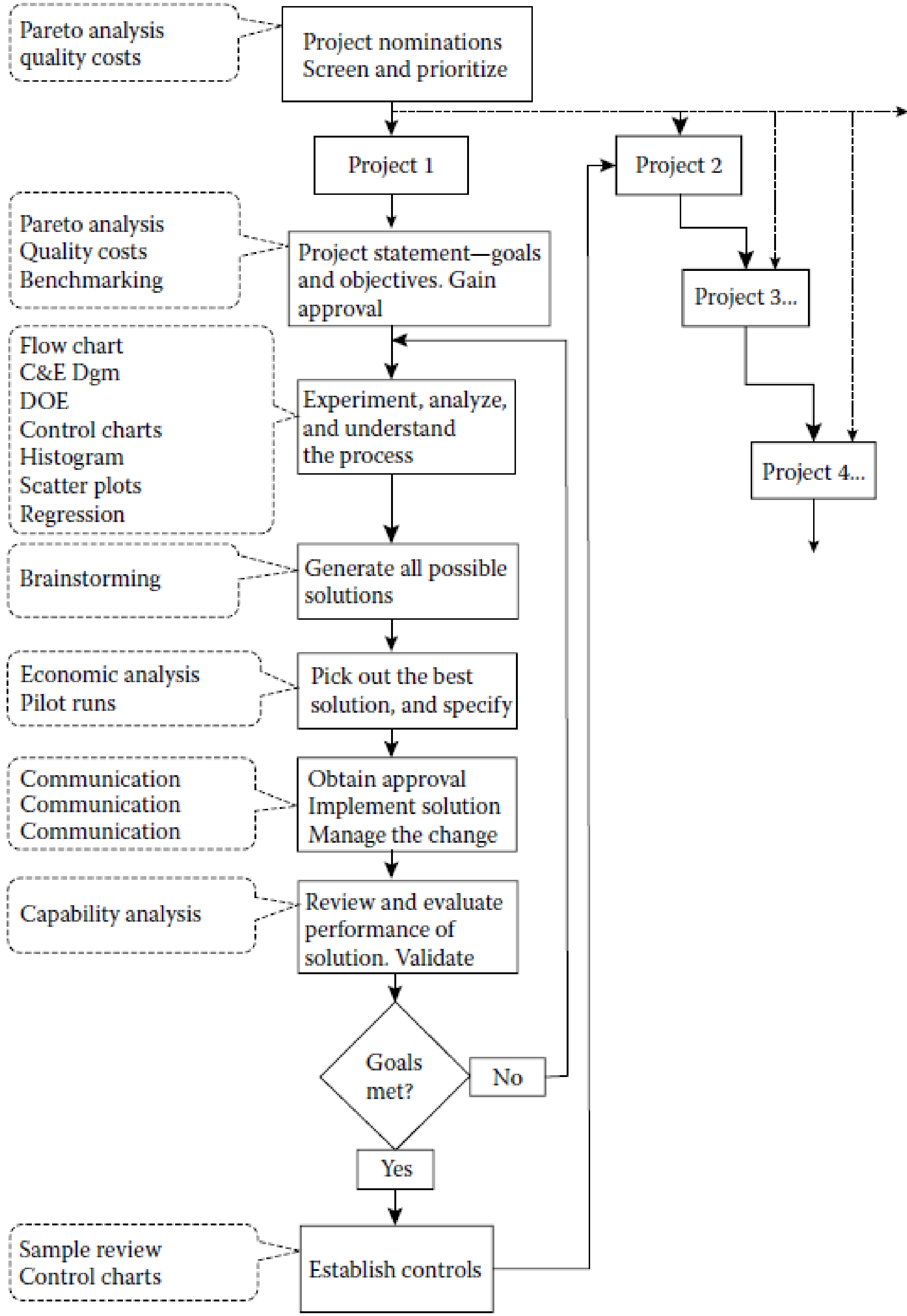
Carry out the change (often a pilot study)

Juran's Breakthrough Sequence

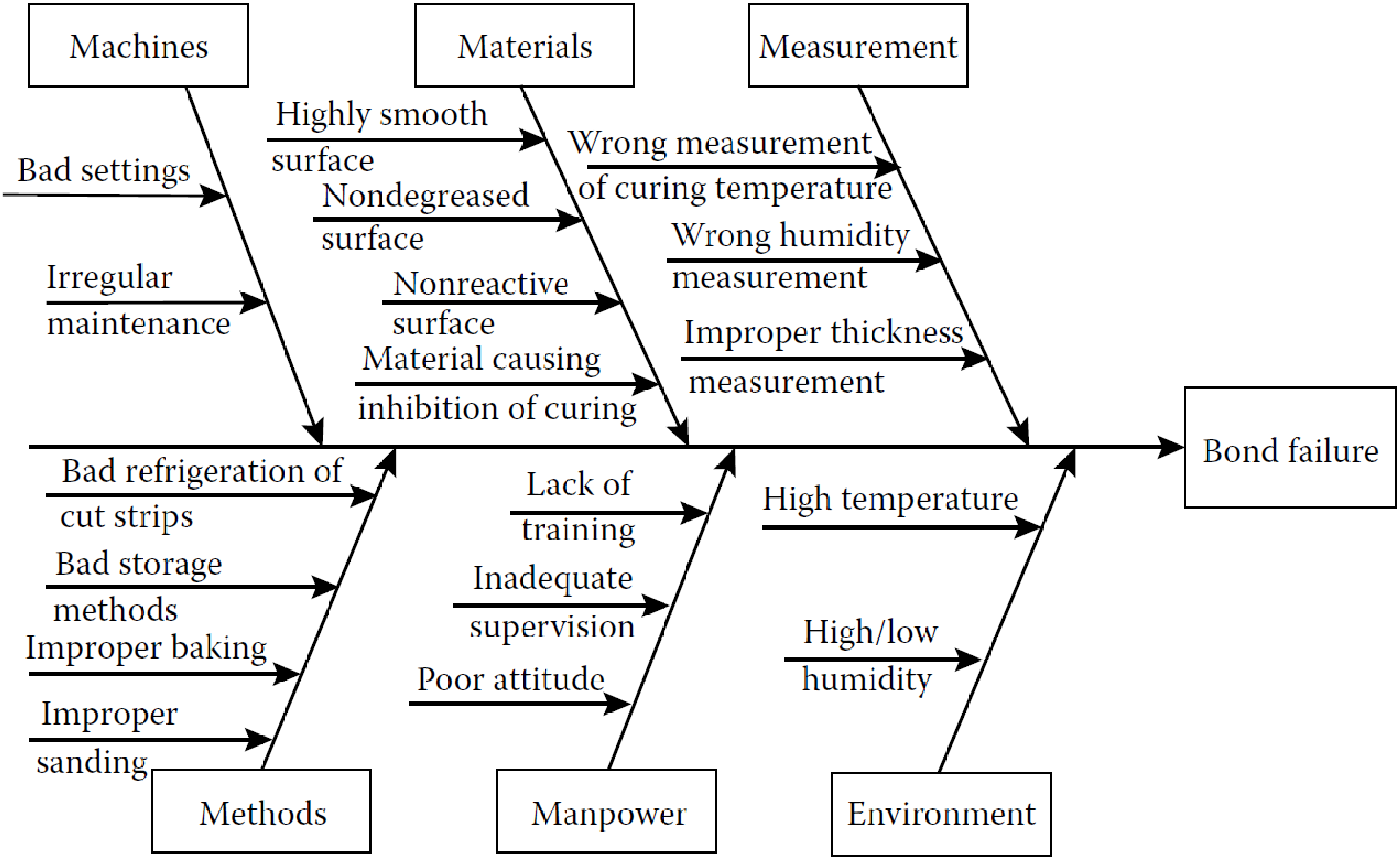
- Establish proof of need
- Identify the project
- Organize a project team
- Make the diagnostic journey
- Make the remedial journey
- Institute controls to hold the gains

Juran's Breakthrough Sequence




Pareto analysis
quality costs
Project nominations
Screen and prioritize
Project 1
Project 2
Project 3...
Project 4...
Pareto analysis
Quality costs
Benchmarking
Project statement—goals and objectives. Gain approval
Flow chart
C&E Dgm
DOE
Control charts
Histogram
Scatter plots
Regression
Experiment, analyze, and understand the process
Brainstorming
Generate all possible solutions
Economic analysis
Pilot runs
Pick out the best solution, and specify
Communication
Communication
Communication
Obtain approval
Implement solution
Manage the change
Capability analysis
Review and evaluate performance of solution. Validate
Goals met?
No
Yes
Sample review
Control charts
Establish controls

Cause-and-Effect Diagram

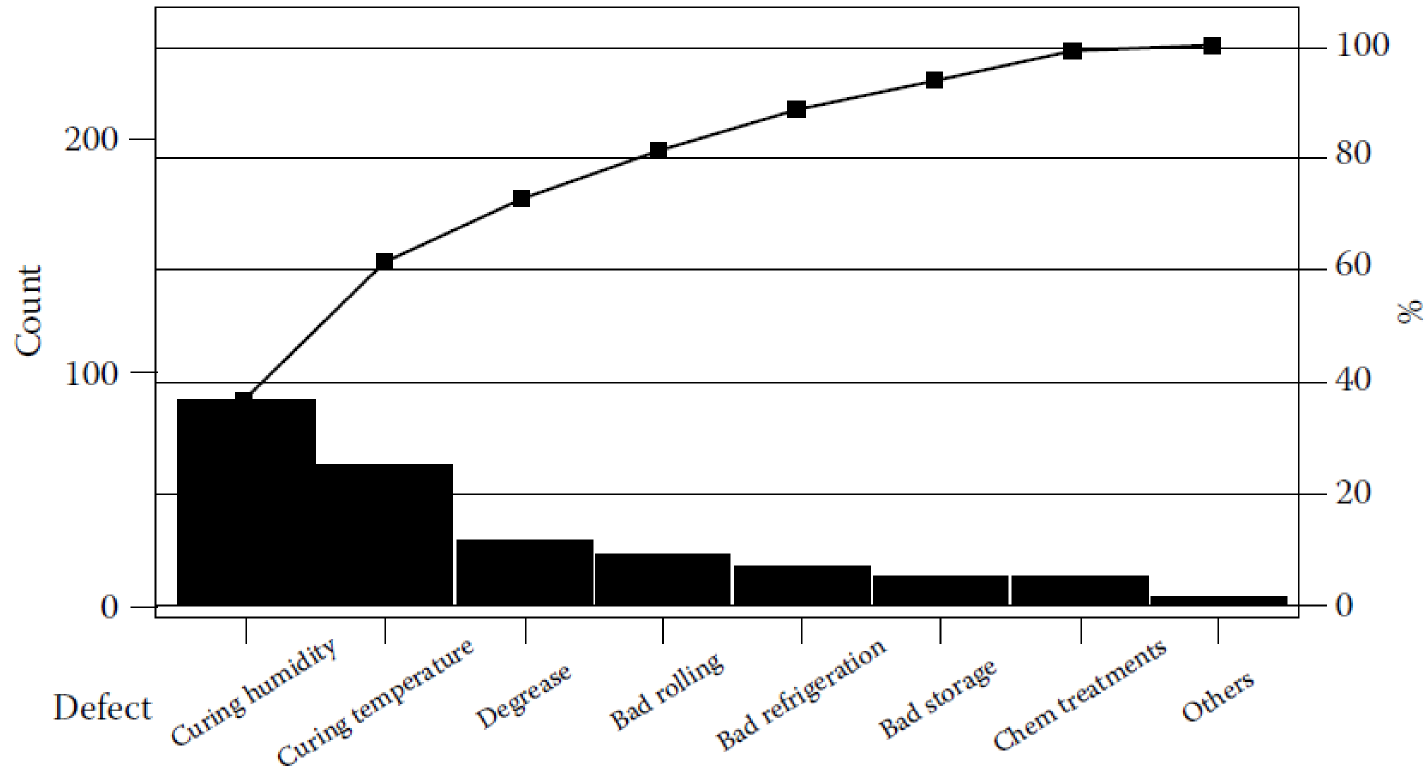


Pareto Analysis

Data on the Causes of Bond Failures

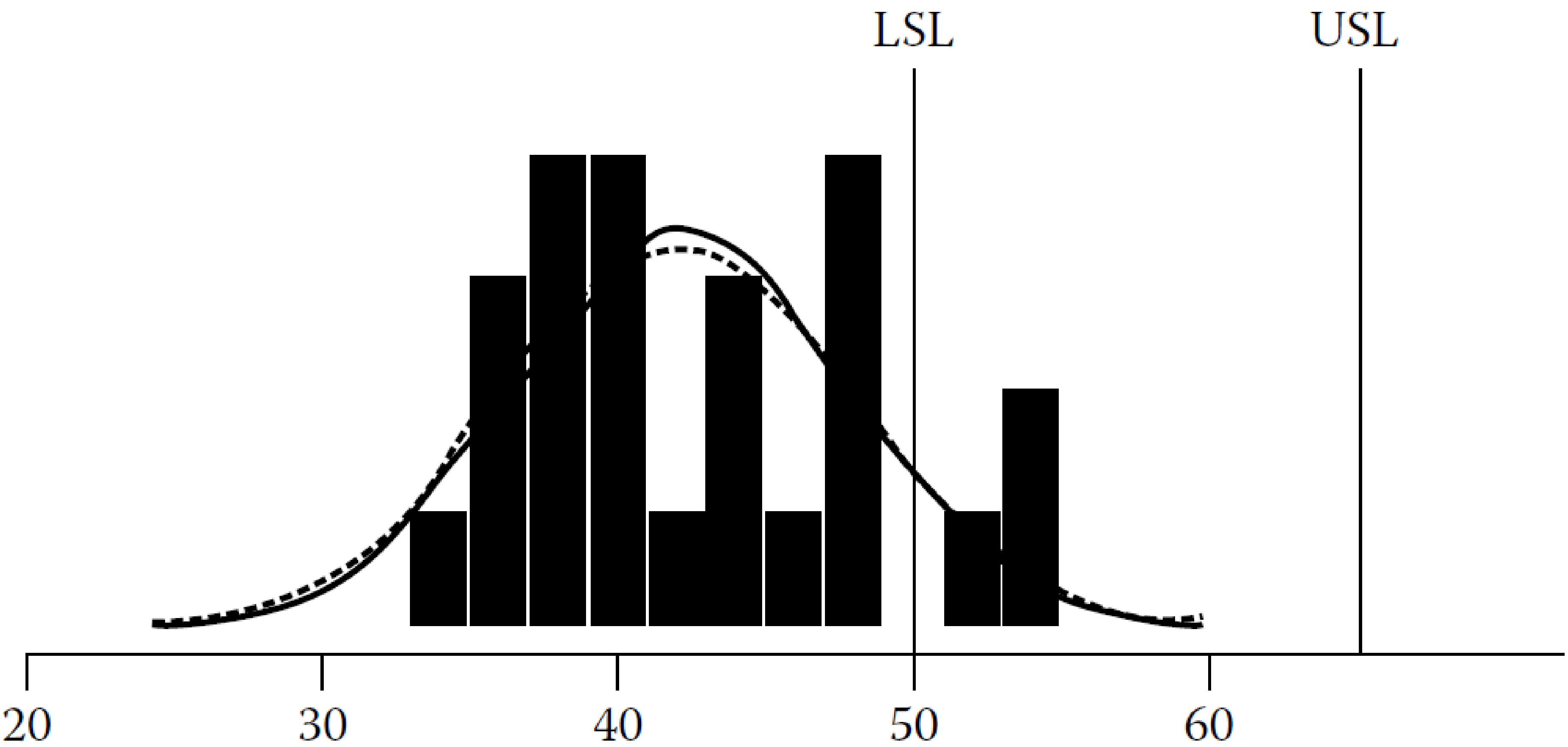
Defect Category	May	June	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	Total
Degrease	3	3	4	2	1	1	2	2	1	7	1	—	27
Chemical treatments	—	—	—	—	—	—	—	—	—	12	—	—	12
Incompatibility	—	—	—	—	—	2	—	1	—	—	—	—	3
Bad storage	—	—	—	1	1	—	—	1	—	9	—	—	12
Bad refrigeration	—	—	3	—	6	4	3	—	—	1	—	—	17
Bad rolling	1	1	—	6	2	7	1	—	—	2	1	—	21
Curing temp.	4	3	4	—	—	6	—	6	5	3	16	12	59
Curing humidity	6	—	3	4	—	4	—	13	13	14	8	22	87
Total	14	7	14	13	10	24	6	23	19	48	26	34	238

Pareto Analysis



Defect	Curing humidity	Curing temperature	Degrease	Bad rolling	Bad refrigeration	Bad storage	Chem treatments	Others
Count	87	59	27	21	17	12	12	3
%	36.6	24.8	11.3	8.8	7.1	5.0	5.0	1.3
Cumulative %	36.6	61.3	72.7	81.5	88.7	93.7	98.7	100.0

Histogram

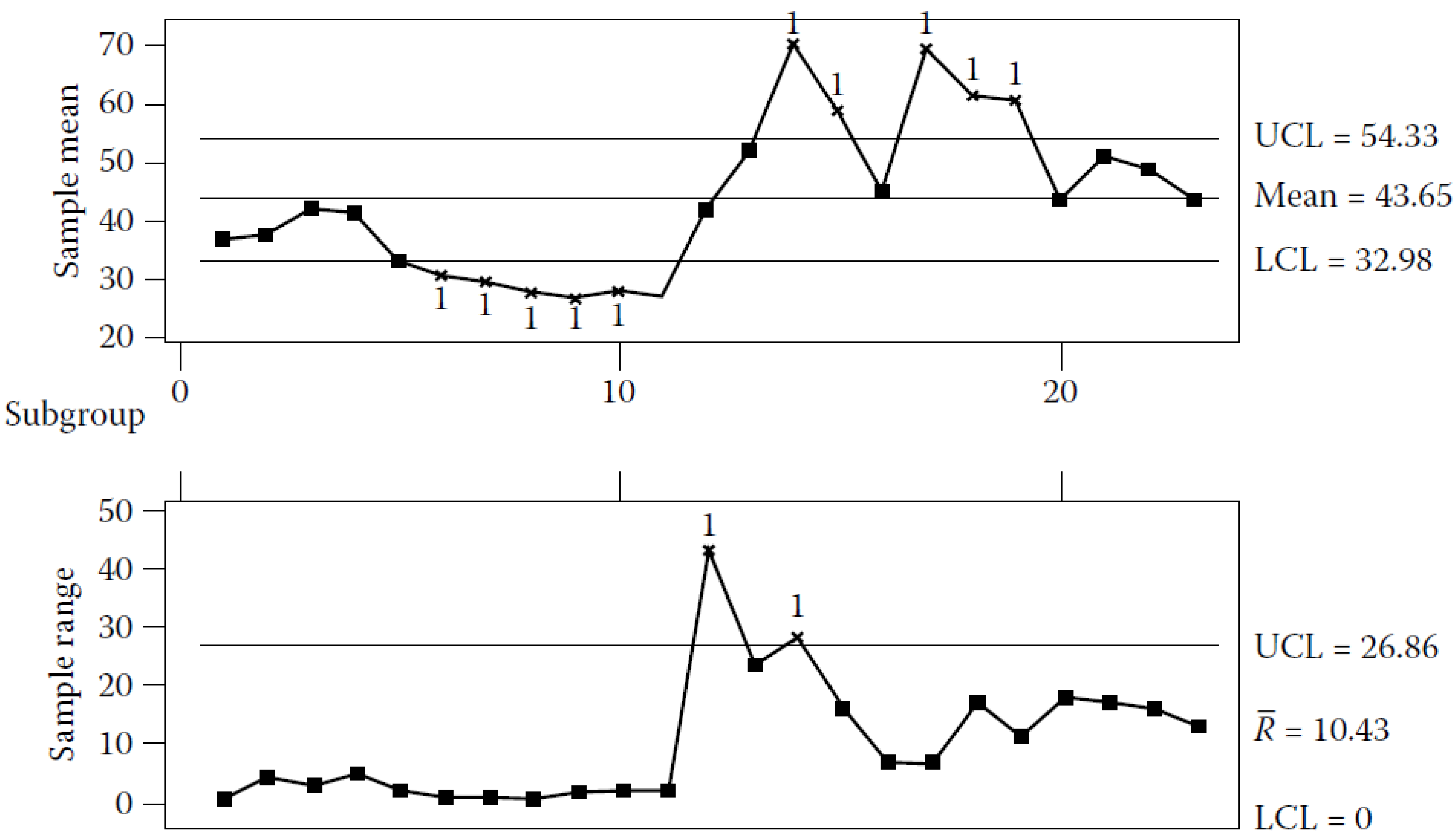


Control Chart

Curing Chamber Humidity in Bond Failure Problem

Subgroup/ Week 1	Humidity %	Subgroup/ Week 2	Humidity %	Subgroup/ Week 3	Humidity %	Subgroup/ Week 3	Humidity %
1	37	8	28	15	50	22	54
	36		27		66		38
	37		28		61		53
2	35	9	26	16	40	23	35
	38		26		47		48
	39		28		47		47
3	40	10	27	17	68		
	43		28		74		
	43		29		67		
4	44	11	26	18	50		
	41		27		67		
	39		28		67		
5	34	12	27	19	64		
	32		28		64		
	33		70		53		
6	30	13	66	20	33		
	30		47		51		
	31		43		46		
7	29	14	53	21	43		
	29		76		50		
	30		81		60		

Control Chart



Scatter Plot